

Research Paper

Experimental investigation on the performance of a Solar-Ocean Thermal Energy Conversion system based on the Organic Rankine Cycle

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ABSTRACT

To improve the performance of the Ocean Thermal Energy Conversion (OTEC) system based on the Organic Rankine Cycle (ORC), a solar-assisted cycle is added, and the power generation experimental setup for the Solar-Ocean Thermal Energy Conversion (S-OTEC) system has been constructed. Through energy and exergy analysis, the effects of solar hot water temperature and working fluid flow rate on the components and system performance are investigated, and the performance differences between S-OTEC and OTEC at the same seawater temperature and working fluid flow rate are compared. The results show that: in the S-OTEC system, as the solar hot water temperature is increased from 54 °C to 72 °C, the isentropic efficiency of the working fluid pump and expander is improved, but the heat transfer coefficient of the heat exchanger is decreased. Furthermore, the thermal efficiency, power generation efficiency, and exergy efficiency increase in the ranges of 2.02–3.26 %, 2.93–3.15 %, and 37.93–41.01 %, respectively. When the working fluid flow rate is increased from 124.33 kg/h to 216.73 kg/h, the isentropic efficiency of the working fluid pump and expander as well as the heat transfer coefficient of the heat exchanger are improved. Additionally, the thermal efficiency, power generation efficiency and exergy efficiency increase in the ranges of 2.05–2.75 %, 2.46–2.97 % and 38.01–48.13 %, respectively. What's more, the thermal efficiency, power generation efficiency and exergy efficiency of the S-OTEC system can be increased by 154.32 %, 39.33 % and 27.87 % respectively compared to the OTEC system.

1. Introduction

The rapid growth of the world's population and the accelerated development of industrialisation have led to a significant increase in energy demand of 2.4 % per year over the past few decades [1], resulting in the depletion of fossil fuels and increasingly serious environmental pollution, which need to be urgently addressed. Under the proposal of “carbon peak” and “carbon neutrality” in the world, it has become the common responsibility of all countries to develop renewable energy, improve the energy structure, solve the energy shortage and protect the ecological environment. As we all know, as the largest solar energy collector on earth, the ocean absorbs about 70 % of the solar energy reaching the earth's surface [2], and simultaneously converts it into various forms of renewable energy, such as tidal energy, wave energy, current energy, and ocean thermal energy, etc. Compared with other types of ocean energy, ocean thermal energy is considered to be a promising ocean energy resource due to its large storage capacity (4.4×10^{16} kWh/a) [3] and relative stability with weather changes [4,5].

To make efficient use of the ocean thermal energy, scientists have

proposed an Ocean Thermal Energy Conversion (OTEC) system based on different thermal cycles. Their working principle is to use the temperature difference between warm surface seawater and cold deep seawater to generate electricity. OTEC systems can be classified into three categories based on differences in cycle materials and structures: open system, closed system, and hybrid system [6,7]. The closed system has become the most recommended option in current research due to its promising large-scale application and easier miniaturisation [8–10], and using the Organic Rankine Cycle (ORC) as the thermal cycle [11]. However, even in the tropics, the maximum temperature difference available for an OTEC system is only about 20 °C–26 °C [12,13], which is much lower than that of a conventional ORC in waste heat recovery, resulting in a very low thermal efficiency of 1.5–3 % [14]. To overcome this difficulty, researchers have improved the OTEC system in three main ways, including system configuration, working fluid selection and component selection.

Increasing the temperature difference of an OTEC system by integrating other energy sources is a well-performing method of improving the system configuration. Solar energy, as a low-grade renewable energy source, has become the preferred energy source for supporting OTEC

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Nomenclature		Subscript	
<i>Symbol</i>		0	Environment state
A	Area (m^2)	1–16,i	state points
C_P	Isobaric specific heat capacity ($\text{kJ}/(\text{kg}\cdot\text{K})$)	con	Condenser
E_X	Exergy (kW)	ele	Electricity
E_{X_D}	Exergy destruction (kW)	exp	Expander
h	Specific enthalpy (kJ/kg)	eva	Evaporator
m	Mass flow rate (kg/s)	in	Inlet
P	Pressure (MPa)	is	Isentropic
p_{rat}	Pressure ratio	net	Net output
Q	Heat transfer (kW)	out	Outlet
s	Specific entropy ($\text{kJ}/(\text{kg}\cdot\text{K})$)	pump	Pump
T	Temperature ($^{\circ}\text{C}$)	rat	Ratio
U	heat transfer coefficient ($\text{W}/(\text{m}^2\cdot\text{K})$)	sat	Saturated
W	Power or power consumption (kW)	sup	Superheater
X	Measured variables	super	Superheating
Y	Derived variables	th	Thermal
<i>Greek letter</i>		wf	Working fluid
Δ	Difference	sys	System
η_{is}	Isentropic efficiency (%)	<i>Acronym</i>	
η_{th}	Thermal efficiency (%)	GWP	Global Warming Potential
η_{ele}	Power generation efficiency (%)	LMTD	logarithmic mean temperature difference
η_{ex}	Exergy efficiency (%)	ODP	Ozone Depletion Potential
ω	exergy loss ratio	ORC	Organic Rankine Cycle
		OTEC	Ocean Thermal Energy Conversion
		S-OTEC	Solar-Ocean Thermal Energy Conversion

systems due to its easy accessibility and complementary characteristics with ocean thermal energy [15]. Initially, researchers introduced solar collectors to preheat warm surface seawater in OTEC systems to increase the input temperature of the heat source. For example, Yamada et al. [16] used numerical simulations to investigate the feasibility of using solar energy to preheat warm seawater in an OTEC system. The results showed that when the temperature of the warm seawater was increased by 20 K, the corresponding net thermal efficiency of the system increased by 1.5 times. Subsequently, Yuan et al. [17] carried out a theoretical study for an OTEC system with the same structure. The results showed that when the seawater temperature at the collector outlet reached 353 K, the power generation efficiency of the system could reach 2.27 %. In addition, scholars in recent studies have proposed using solar energy to directly heating the working fluid. Aydin et al. [18] compared two methods of preheating warm seawater or superheating the working fluid using solar collectors. The results showed that the net power generation increased by 20–25 % in both cases. However, the preheating case corresponds to lower thermal efficiency. In contrast, the superheating case not only increased the thermal efficiency of the system from 1.9 % to 3 %, but also significantly reduced the collector area required. Therefore, using solar collectors to superheat the working fluid is a better way to improve the thermal efficiency of OTEC systems than preheating warm seawater. Recently, Huo et al. [19] also added a solar superheated working fluid sub-cycle to the OTEC system, investigating the effect of different parameters on system performance to determine the optimal operating conditions. The results showed that when the heat transfer fluid in the solar collector was reduced from 150 °C to 97.5 °C, the maximum primary energy saving rate and exergy efficiency of the system reached 9.99 % and 11.94 %, respectively. In this study, it is determined that the solar collector superheated working fluid is selected to improve the system performance.

The characteristics of the working fluid can have a major impact on component design and system stability, efficiency and cost. In addition to thermal performance, the environmental impact of the working fluid must generally be considered, with specific requirements for zero Ozone

Depletion Potential (ODP) and low Global Warming Potential (GWP). Recently, some scientists have advocated the selection of zeotropic mixtures as working fluid for OTEC system [13,14,20–22]. They pointed out that mixtures lead to temperature slip during the phase change process, resulting in a better temperature match between the working fluid and seawater, which is conducive to the reduction of irreversible losses in the heat transfer process. However, temperature slip reduces the average heat transfer temperature difference or heat transfer coefficient, which naturally increases the heat transfer area and therefore the investment cost [20,23,24]. Additionally, the different velocities of the liquid and gaseous phases of the working fluid during the phase change process lead to compositional shifts that cause the ORC to deviate from the designed operating conditions, thus reducing the thermodynamic performance of the ORC [25,26]. The continuous phase change process changes the compositional ratios of the mixture, making it difficult to calculate the thermal parameters for a given state, from which it is difficult to make more accurate and comprehensive calculations, leading to an increase in the percentage of error in the calculation of thermophysical properties [23,27]. Despite the thermodynamic and economic problems associated with non-zeotropic variable mixtures, it is possible to extend the choice of working fluids. In addition, for pure working fluids, R717 is still a widely used working fluid for OTEC systems due to its excellent thermodynamic properties [28–30]. However, considering that it is not only toxic and flammable, but also associated with the possibility of explosion in certain environments, it is not applicable in practical demonstration experiments. Therefore, researchers have never ceased to search for a more suitable pure fluid as an alternative to R717. Fiaschi et al. [31] evaluated the performance of different working fluids in a small-scale ORC system. It was found that the fluid like R134a with high density as well as low volume expansion ratio can enable the expander to achieve favourable working efficiency and ensure high thermal performance of the system. However, it is not yet known which pure working fluid is best suited for OTEC-ORC.

The proper selection of the components for an OTEC system is the key to improving the thermal efficiency, especially the type of expander.

This is because the type of expander is often influenced by factors such as the heat source temperature, the type of working fluid and the power generation requirements [32]. Declaye et al. [33] and Zhao et al. [34] have successively classified and summarised the expansion devices in cryogenic ORC systems. They all pointed out that volumetric expanders are more suitable for micro ORC systems, especially scroll expanders. More importantly, the data emphasises that the micro-ORCs mentioned above are essentially defined as cycles with an expected power of around 0.35–7.5 kW [35]. The popularity of scroll expanders for small-scale ORC systems is due to their low number of moving parts, compactness, low cost, low noise and reliable performance [36,37]. In addition, for the scroll expander, its efficiency is often determined by factors such as the shape of the blades, the fluid properties and the outlet degree of expansion [11]. In recent years, researchers have also carried out optimisation studies of expanders using fluid dynamic analysis and numerical simulation. However, the efficiency of the improved scroll expander is relatively low [36], which makes the performance improvement of the small-scale OTEC system unsatisfactory [38]. Therefore, it is more important to pay attention to the selection of the expander in the small-scale OTEC experimental system rather than how to improve the expander structure. Since the heat transfer temperature difference in this study is small, the heat transfer power of the heat source is not higher than 20 kW, and the expected power is only 1 kW, the scroll expander is selected as the thermoelectric converter.

In summary, the performance of the OTEC system can be effectively improved by selecting a better thermodynamic working fluid and components with higher working efficiency, and by adding a solar-assisted circulation structure. However, most of the current studies to improve OTEC systems using the above methods are limited to simulation analysis [19,38,39]. The lack of relevant experimental studies makes it difficult to provide strong data support for the construction of actual large-scale power generation. Therefore, in order to empirically measure the performance enhancement effect of Solar-Ocean Thermal

Energy Conversion (S-OTEC) system, this study aims to construct an experimental setup for a 1 kW S-OTEC thermal system. R134a was used as the working fluid, and a scroll expander was used as the output device of the system, while solar collector superheated working fluid was used. Through energy analysis and exergy analysis, the influence of solar hot water temperature and working fluid flow on the performance of the S-OTEC system is studied and the performance difference between the S-OTEC system and the OTEC system is compared.

The paper is structured as follows: Section 2 describes the experimental system, Section 3 summarises the thermodynamic analysis methodology, Section 4 presents the results and discussion, and Section 5 provides conclusions. The experimental results provide a reference for the control and operation of the constructed system.

2. Description of the experimental system

The schematic diagram of the 1 kW S-OTEC system is shown in Fig. 1, which consisting of three circuits: the heating circuit (red line), the power generation circuit (black line) and the cooling circuit (blue line). The experimental setup allows for two different operating modes (OTEC and S-OTEC) by controlling whether the solar collector is connected to the system or not. The experimental setup and main components are shown in Fig. 2.

2.1. Working circuits description

2.1.1. Heating circuit

The tap water is used as the heat transfer medium in the heating circuit. It is heated to different temperature by electric or solar heating to act as the heat source for the system.

a) Electric heating cycle: an electric heater with a maximum heating power of 20 kW provides hot water of 25–31 °C for simulating surface warm seawater. The hot water is transported through an evaporator to

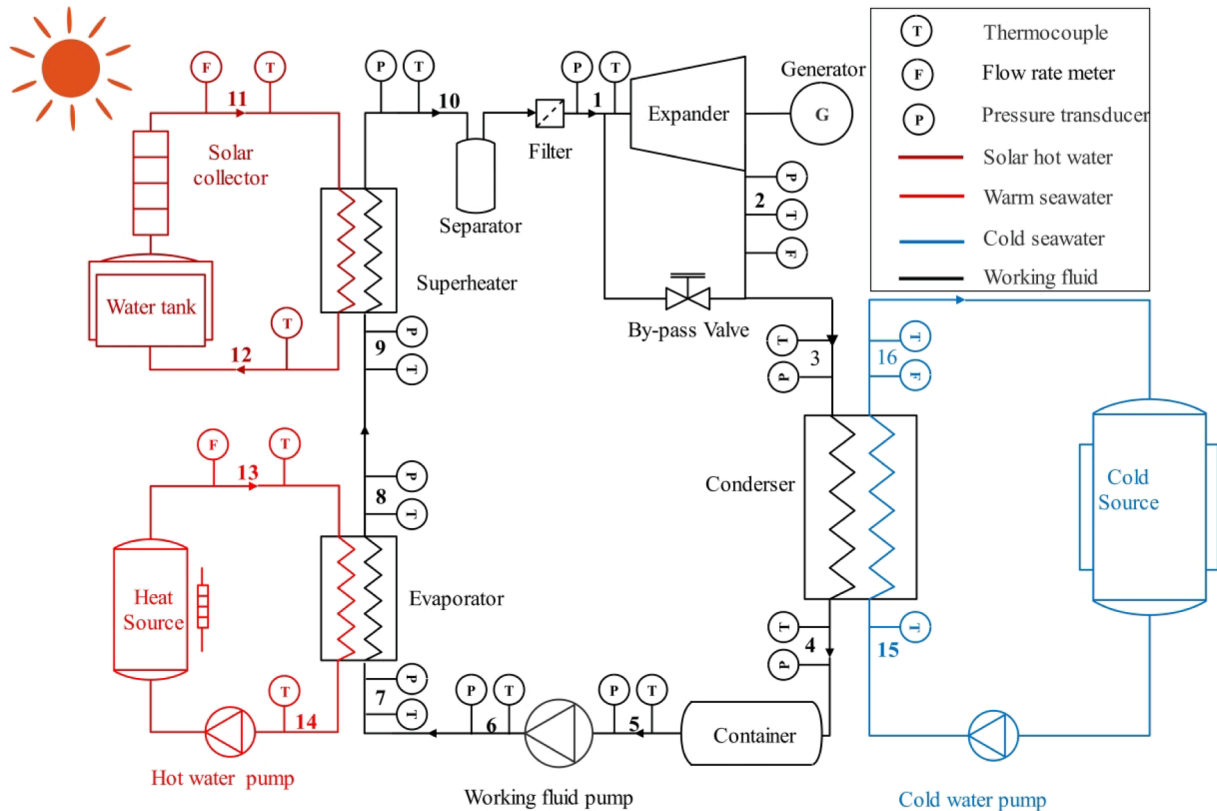
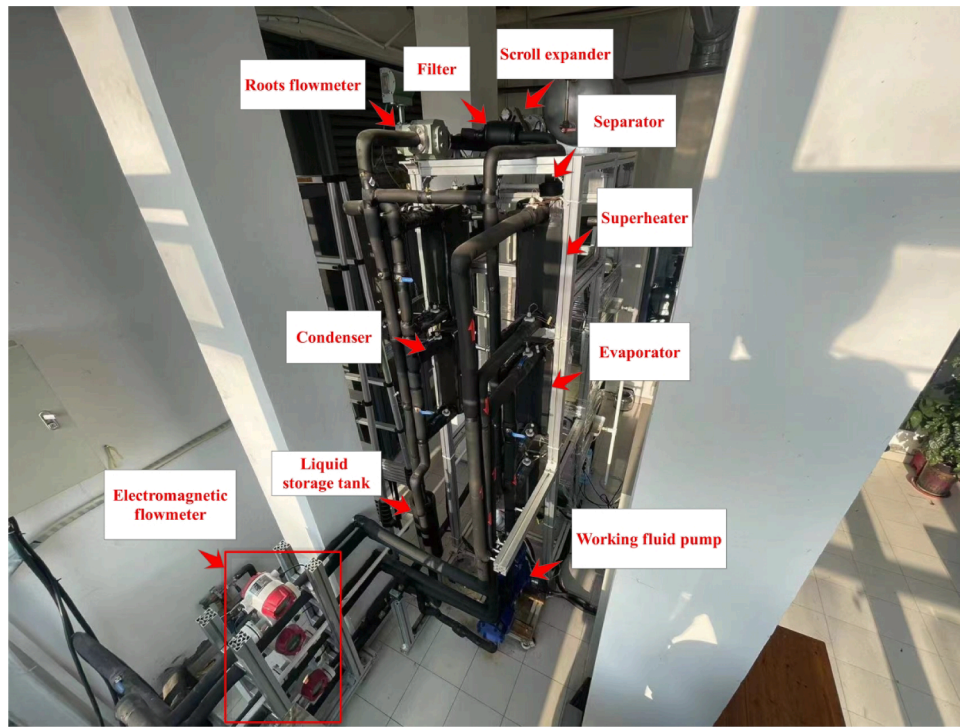


Fig. 1. Schematic diagram of a 1 kW S-OTEC system.



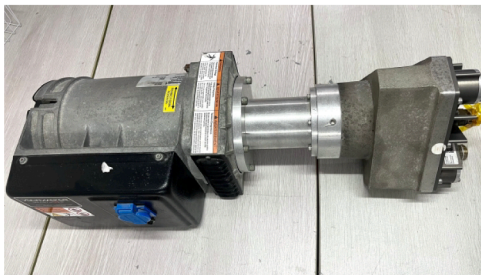
(a)



(b)



(c)



(d)



(e)

Fig. 2. Photographs of the S-OTEC experimental system, (a) S-OTEC experimental setup.

deliver heat to the power generation circuit and back to the electrically heated water tank. The hot water temperature is regulated by a PID controller and the flow rate is regulated by variable frequency.

b) Solar heating cycle: In S-OTEC mode, solar-absorbed water is pumped into the superheater to heat the circulating working fluid to a superheated state and return to the hot water storage tank. The flow rate of circulating water is regulated by variable frequency. The two groups

of vacuum tube collectors selected for the experimental system are connected in parallel, in which each group contains 100 vacuum tubes. The relevant parameters of the vacuum tubes are as follows: the type is vacuum titanium three-high microwave collector tube, the outer diameter is 58 mm, the inner diameter is 44 mm, and the tube length is 1800 mm.

2.1.2. Power generation circuit

The power generation circuit consists of the main components (evaporator, superheater, condenser, expander, and working fluid pump) and other auxiliary components (gas–liquid separator, filter, liquid storage tank, etc.). Although alternative fluids with lower GWP are expected to replace hydrofluorocarbons like R134a in the future, a substantial number of existing ORC still select R134a as working fluid [40]. Additionally, R134a not only has the environmental characteristics of zero ODP and low GWP, but also its thermodynamic characteristics such as low boiling point and high decomposition temperature are very suitable for the working conditions of the OTEC system [41]. Simultaneously, its non-toxic and non-flammable characteristics also very much satisfy the safety guideline of the test bench [42]. Therefore, R134a is selected as the working fluid for the power generation circuit, taking into account thermodynamic, environmental and safety characteristics. Further properties of R134a are shown in Table 1.

In the S-OTEC system, R134a is driven by the working fluid pump into the evaporator to absorb heat from warm seawater, and then heated by solar hot water in the superheater to become superheated vapor. The superheated R134a vapor passes through the separator and filter, and enters the expander to push the impeller to rotate and drive the generator for electricity generation. The low-pressure exhaust vapor enters the condenser to be cooled into subcooling liquid and flows into the storage tank, where it enters the working fluid pump to start a new cycle. It should be noted that when the solar heating cycle is not connected to the system, the operating mode of the experimental setup is transformed to OTEC. The detailed information on the main components of the power generation circuit are shown in Table 2, Table 3, Table 4 and Table 5 respectively.

2.1.3. Cooling circuit

The cooling circuit uses tap water as the heat transfer medium, and the temperature range of the water is controlled by the refrigeration unit in the range of 4–8 °C to simulate deep cold seawater. The cold seawater is pumped into the condenser to cool the exhaust vapor into the sub-cooled liquid, which is then returned to the refrigeration unit to be compressed and cooled for the next cycle. The temperature and flow rate of the cold seawater are controlled and stabilised by PID controller.

2.2. Measuring equipment

As shown in Fig. 1, thermocouples and pressure sensors are located in the import and export of key equipment to measure temperature and pressure. Flow meters are located in the heating, power generation circuit and cooling circuits to measure the flow of process material and circulating water. In addition, the power meter monitors power generation in real time. The accuracy of all the above meters is shown in Table 5. The measurement data connected to the NI data acquisition system can all be displayed on the computer control interface and stored in the data acquisition software. During post-processing of the data, some of the thermophysical parameters (enthalpy, entropy, density, etc.) are retrieved in conjunction with REFPROP 9.1 to complete the thermodynamic calculation and analysis process of the system.

Table 1
Main characteristic parameters of R134a [42,43].

Properties	Unit	Value
Molar mass	g/mol	102.3
Boiling temperature	°C	−26.1
Decomposition temperature	°C	368–389
Critical temperature	°C	101.1
Critical pressure	MPa	4.06
Toxicity	/	A
Flammability	/	A1
ODP	/	0
GWP	/	1430

Table 2

Main parameters of the Evaporator/ Superheater.

Parameters	Value
Type	Brazed plate heat exchanger
Tube material	Stainless steel
Plates number	40
Temperature range	−196–200 °C
Maximum pressure	3 MPa
Heat transfer area	2.09 m ²

Table 3

Main parameters of the Condenser.

Parameters	Value
Type	Brazed plate heat exchanger
Tube material	Stainless steel
Plates number	40
Temperature range	−196–225 °C
Maximum pressure	4 MPa
Heat transfer area	6.54 m ²

Table 4

Main parameters of the Expander.

Parameters	Value
Type	Semi-hermetic scroll expander
Swept volume	14.5 cm ³ /rpm
Built-in volume ratio	3.5
Maximum inlet pressure	1.38 MPa
Maximum inlet temperature	175 °C
Power output	1 kW

Table 5

Main parameters of the Working fluid pump.

Parameters	Value
Type	Hydraulic diaphragm metering pump
Volume flow range	0–200 L/h
Power consumption	0.55 kW
Maximum outlet pressure	1.6 MPa
Minimum inlet temperature	−30 °C

The parameters in this study can be divided into directly measured parameters and derived parameters. The directly measured variables include water flow rate, water temperature, working fluid temperature, working fluid pressure, working fluid flow rate and electrical power. In this case, the uncertainty and the relative uncertainty of the derived parameters are calculated according to eq. 4 and eq.5, respectively [44]. Table 6 summarises the accuracy of the measuring instrumentation and the relative uncertainty of the main parameters.

$$\Delta Y = \sqrt{\sum_i \left(\frac{\partial Y}{\partial X_i} \right)^2 \Delta X_i^2} \quad (1)$$

$$R_Y = \Delta Y / Y \quad (2)$$

Where, $\Delta Y / Y$ is the sensitivity coefficient of the parameter Y relative to X_i , ΔX_i is the uncertainty of X_i , ΔY is the uncertainty of Y . R_Y is the relative uncertainty of the parameter Y .

3. Thermodynamic analysis method

3.1. Energy analysis model

The T - S diagram of the S-OTEC system is shown in Fig. 3. (The two measurement points inside and outside the parentheses in the T - S

Table 6

The measuring equipment accuracy and the main parameters relative uncertainty.

Parameter	Device type	Range	Accuracy
Temperature (°C)	T-Type Thermocouple	0–260	±0.5 °C
Pressure (MPa)	Pressure transducer	0–2.5	±0.1 % FS
Water volume flow (m ³ /h)	Electromagnetic flowmeter	0.32–3.2	±0.5 % FS
R134a volume flow (m ³ /h)	Roots flowmeter	2–14	±0.2 % FS
Power generation (W)	Power meter	0–1000	±0.5 % FS
Analysis result			Relative uncertainty
Output power (W)			±5.43 %
Pump power consumption (W)			±6.17 %
Net output power (W)			±5.67 %
Total heat input (kW)			±2.40 %
Thermal efficiency (%)			±5.73 %
Exergy efficiency (%)			±6.78 %
Power generation efficiency (%)			±5.94 %

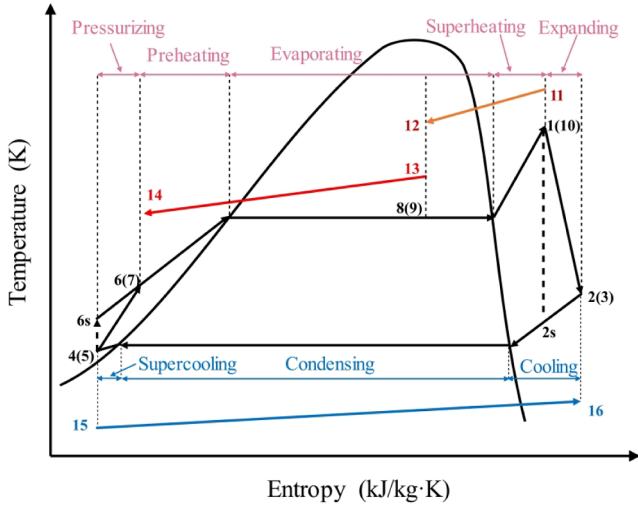
**Fig. 3.** T-s plot of the S-OTEC system.

diagram have the same state, and the positions of all the measurement points correspond to the schematic diagram.) The specific change process is as follows:

- 1 → 2: Working fluid is expanding in the expander.
- 2 → 4: Working fluid is condensed in the condenser.
- 4 → 7: Working fluid is compressed by the working fluid pump.
- 7 → 8: Working fluid is preheated and evaporated in the evaporator.
- 8 → 10: Working fluid is evaporated and superheated in the superheater.
- 11 → 13: Solar hot water releases heat to working fluid in the superheater.
- 13 → 14: Surface warm seawater releases heat to working fluid in the evaporator.
- 15 → 16: Deep cold seawater receives heat from working fluid in the condenser.

In order to complete the thermodynamic analysis process of the system, some of the thermodynamic equations listed under the following conditional assumptions are shown in Table 7.

- 1) Heat loss and pressure loss in the pipeline are ignored.
- 2) The data selected for the calculations are obtained under steady state conditions.
- 3) The flow rate of all fluids in the pipes and components is uniform and constant.

Table 7

Mathematical models for experiment results analysis.

Components	Contents	Models	Eq.
Working fluid	Mass flow rate	m_{wf}	
	Super-heating degree	$\Delta T_{super} = T_{sup,out} - T_{sup,sat}$	(3)
	Heat absorbed in evaporator	$Q_{eva, wf} = m_{wf}(h_8 - h_7)$	(4)
	Heat absorbed in superheater	$Q_{sup, wf} = m_{wf}(h_{10} - h_9)$	(5)
Evaporator	Heat released in condenser	$Q_{con, wf} = m_{wf}(h_3 - h_4)$	(6)
	Temperature difference	$\Delta T_{eva} = T_{13} - T_{14}$	(7)
	Heat released in evaporator	$Q_{eva, ww} = C_p m_{13} \Delta T_{eva}$	(8)
	Logarithmic mean temperature difference	$LMTD_{eva} = \frac{(T_{13} - T_8) - (T_{14} - T_7)}{\ln[(T_{13} - T_8)/(T_{14} - T_7)]}$	(9)
Superheater	Heat transfer coefficient	$U_{eva} = Q_{eva, wf} / (LMTD_{eva} \cdot A_{eva})$	(10)
	Temperature difference	$\Delta T_{sup} = T_{11} - T_{12}$	(11)
	Heat released in superheater	$Q_{sup, sw} = C_p m_{11} \Delta T_{sup}$	(12)
	Logarithmic mean temperature difference	$LMTD_{sup} = \frac{(T_{11} - T_{10}) - (T_{12} - T_9)}{\ln[(T_{11} - T_{10})/(T_{12} - T_9)]}$	(13)
Expander	Heat transfer coefficient	$U_{sup} = Q_{sup, wf} / (LMTD_{sup} \cdot A_{sup})$	(14)
	Pressure difference	$\Delta p_{exp} = p_1 - p_2$	(15)
	Pressure ratio	$p_{rat} = p_1 / p_2$	(16)
	Actual enthalpy difference	$\Delta h_{exp} = h_1 - h_2$	(17)
Condenser	Ideal enthalpy difference	$\Delta h_{exp, is} = h_1 - h_{2, is}$	(18)
	Isentropic efficiency	$\eta_{exp, is} = (h_1 - h_2) / (h_1 - h_{2, is})$	(19)
	Output power	$W_{exp} = m_{wf} \Delta h = m_{wf}(h_1 - h_2)$	(20)
	Temperature difference	$\Delta T_{con} = T_{16} - T_{15}$	(21)
Working fluid pump	Heat absorbed in condenser	$Q_{con, cw} = C_p m_{15} \Delta T_{con}$	(22)
	Logarithmic mean temperature difference	$LMTD_{con} = \frac{(T_3 - T_{16}) - (T_4 - T_{15})}{\ln[(T_3 - T_{16})/(T_4 - T_{15})]}$	(23)
	Heat transfer coefficient	$U_{con} = Q_{con} / (LMTD_{con} \cdot A_{con})$	(24)
	Temperature difference	$\Delta T_{pump} = T_6 - T_5$	(25)
S-OTEC system	Pressure difference	$\Delta p_{pump} = p_6 - p_5$	(26)
	Actual enthalpy difference	$\Delta h_{pump} = h_6 - h_5$	(27)
	Ideal enthalpy difference	$\Delta h_{pump, is} = h_6 - h_{5, is}$	(28)
	pump power consumption	$W_{pump} = m_{wf}(h_6 - h_5)$	(29)
S-OTEC system	Isentropic efficiency	$\eta_{pump, is} = \frac{(h_{6, is} - h_5)}{(h_6 - h_5)}$	(30)
	Net output power	$W_{net} = W_{exp} - W_{pump}$	(31)
	Thermal efficiency	$\eta_{th} = W_{net} / (Q_{eva, wf} + Q_{sup, wf})$	(32)
	Power generation efficiency	$\eta_{ele} = W_{ele} / (Q_{eva, wf} + Q_{sup, wf})$	(33)

4) The ambient temperature is 35 °C and the ambient pressure is 101.325 kPa.

3.2. Exergy analysis model

The exergy of material status point (Ex) could be defined as:

$$Ex = m[(h - h_0) - T_0(s - s_0)] \quad (34)$$

Where, h_0 is the specific enthalpy of the reference point, kJ/kg. s_0 is the specific entropy of the reference point, kJ/(kg·K). T_0 is the reference temperature, K. Subscript 0 indicates the reference state (101.325 kPa and 308.15 K).

The component exergy loss ($Ex_{D,i}$) is defined as the difference between the input exergy (Ex_{in}) and the output exergy (Ex_{out}) by the

following equation:

$$Ex_D = Ex_{in} - Ex_{out} \quad (35)$$

Based on the above equation, the exergy loss of each component in the S-OTEC system is derived as shown in Table 8.

In addition, the component exergy loss ratio (ω_i) is defined as the component exergy loss ($Ex_{D,i}$) to the system total energy loss ($Ex_{D,sys}$). The specific defining equation is:

$$\omega_i = \frac{Ex_{D,i}}{Ex_{D,sys}} = \frac{Ex_{D,i}}{Ex_{D,eva} + Ex_{D,sup} + Ex_{D,exp} + Ex_{D,con} + Ex_{D,pump}} \quad (41)$$

The total exergy (Ex_{sys}) obtained by the system is:

$$Ex_{sys} = (Ex_{11} + Ex_{13} + Ex_{15}) - (Ex_{12} + Ex_{14} + Ex_{16}) + W_{pump} + W_{exp} \quad (42)$$

The exergy efficiency of the system (η_{ex}) is:

$$\eta_{ex} = 1 - \frac{Ex_{D,sys}}{Ex_{sys}} \quad (43)$$

4. Results and discussion

4.1. Effect of solar hot water temperature on system performance

In the experiment process, the solar hot water temperature (T_{11}) was varied at 3 °C intervals in the range 54–72 °C. Potential optimisation directions are further analysed by investigating the effects of T_{11} on the components as well as the system performance. The main experimental operating parameters are shown in Table 9.

4.1.1. Performance of the heat exchanger

Fig. 4(a) displays the variation pattern of the outlet pressure of the superheater (P_{10}) and the outlet pressure of the condenser (P_4) with T_{11} . It can be seen that with the increase in T_{11} , P_{10} and P_4 increase monotonically in the ranges of 1.07–1.13 MPa and 0.388–0.396 MPa, respectively. This is because as the temperature of the working fluid increases, its saturation pressure increases, resulting in an increase in the evaporating pressure and condensing pressure. Therefore, P_{10} and P_4 , which represent the evaporating pressure and condensing pressure respectively, show an increasing trend.

Fig. 4(b) shows the variation pattern of working fluid flow rate (m_{wf}) and super-heating degree (ΔT_{Super}) with T_{11} . With the increase in T_{11} , m_{wf} decreases from 163.45 kg/h to 151.18 kg/h and ΔT_{Super} increases from 11.60 °C to 26.40 °C. The reason for the decrease in m_{wf} is that for the same volume flow, the increase in T_{11} decreases the density of the working fluid. In addition, the increase in T_{11} causes the working fluid temperature at the outlet of the superheater increases significantly, but as shown in Fig. 4(a), it can be seen that the increase of P_{10} is only small, which causes the saturation temperature to increase only slightly. Combined with Eq. (7), it can be seen that ΔT_{Super} keeps increasing.

Fig. 5 demonstrates the variation pattern on the heat transfer effect of heat exchangers with T_{11} , including logarithmic mean temperature difference ($LMTD$) and heat transfer coefficient (U). With the increase in T_{11} , $LMTD_{eva}$, $LMTD_{sup}$ and $LMTD_{con}$ monotonically increase in the range of 3.71–5.73 °C, 7.01–14.01 °C and 13.92–18.16 °C, U_{eva} , U_{sup} and U_{con} monotonically decrease in the range of 129.73–77.64 W/(m²·K),

Table 9

Main operating parameters used in the experiment.

Parameters	Value	Unit
Environmental temperature	30	°C
Temperature of warm seawater at evaporator inlet	31	°C
Temperature of cold seawater at condenser inlet	5	°C
Temperature of solar hot water at collector outlet	54–72	°C
The Surface warm seawater flow rate	1.5	m ³ /h
The deep cold seawater flow rate	2.5	m ³ /h
The solar hot water flow rate	1.5	m ³ /h

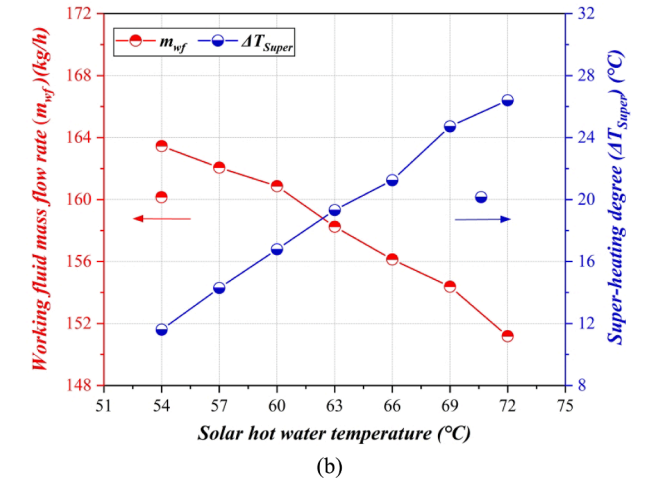
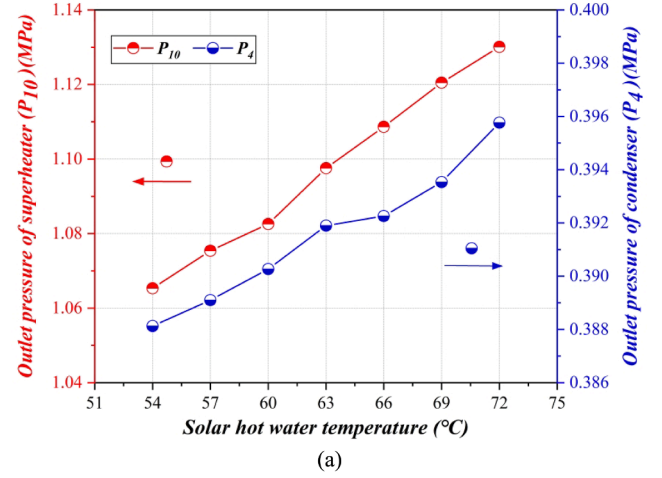


Fig. 4. Variation of the basic parameters of heat exchanger with T_{11} increasing.

474.15–246.08 W/(m²·K) and 118.11–91.43 W/(m²·K), respectively. This means that the weakening of the heat transfer effect in the heat exchanger and increasing the exergy loss of the heat exchangers. Additionally, compared with U_{sup} , U_{eva} and U_{con} have a smaller decrease of 52.09 W/(m²·K) and 26.68 W/(m²·K), respectively, which shows that the superheater has a larger magnitude of exergy loss.

4.1.2. Performance of the working fluid pump

Fig. 6 displays the effect of T_{11} on the performance of the working fluid pump. In Fig. 6(a), as T_{11} increases, the temperature difference of the pump (ΔT_{pump}) decreases from 2.58 °C to 1.78 °C and the Pressure difference (ΔP_{pump}) increases from 0.68 MPa to 0.75 MPa. It is the decrease in ΔT_{pump} as well as the increase in ΔP_{pump} that leads to a decrease in the enthalpy difference of the pump (Δh_{pump}). As shown in the Fig. 6(a), Δh_{pump} decreases from 3.55 kJ/kg to 2.55 kJ/kg.

The isentropic efficiency of the pump ($\eta_{pump,is}$) is shown to increase

Table 8

Component exergy Destruction of the Power generation circuit.

Component	Exergy loss	Eq.
Evaporator	$Ex_{D,eva} = E_{13} - E_{14} + E_7 - E_8$	(36)
Superheater	$Ex_{D,sup} = E_{11} - E_{12} + E_9 - E_{10}$	(37)
Expander	$Ex_{D,exp} = E_1 - E_2 - W_{exp}$	(38)
Condenser	$Ex_{D,con} = E_{15} - E_{16} + E_3 - E_4$	(39)
Working fluid pump	$Ex_{D,pump} = E_5 - E_6 + W_{pump}$	(40)

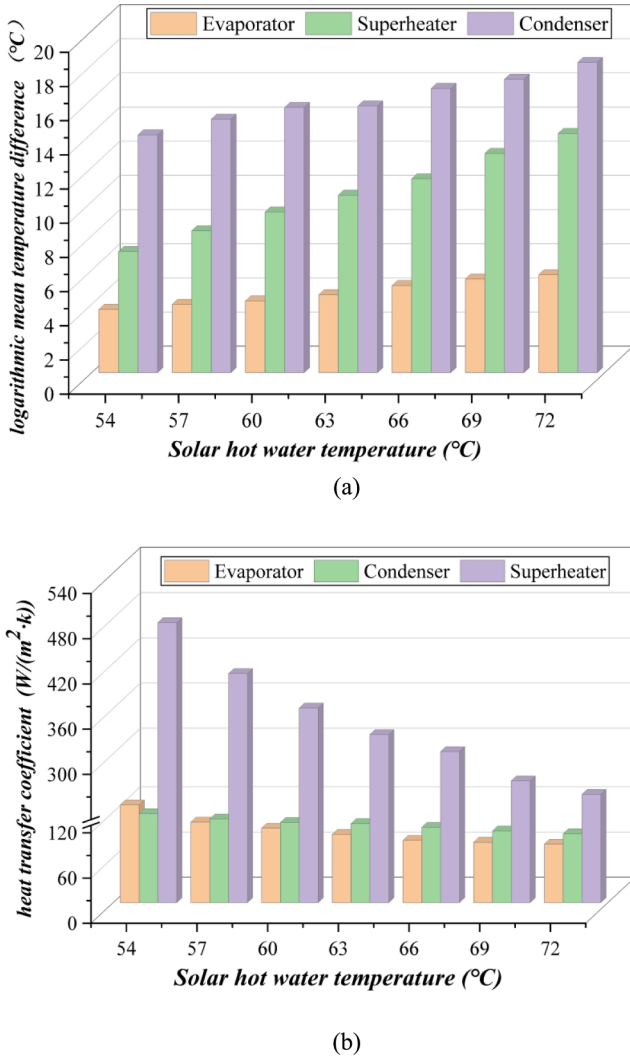


Fig. 5. Variation of the heat transfer effect of heat exchangers with T_{11} increasing.

from 15.53 % to 23.89 % in Fig. 6(b). This is due to the fact that the ideal enthalpy difference of the pump ($\Delta h_{\text{pump, is}}$) slightly increases as Δh_{pump} decreases. In addition, the pump power consumption (W_{pump}) decreases monotonically in the range of 147.63–111.08 W as T_{11} increases. This is due to the fact that both m_{wf} and Δh_{pump} continue to decrease as T_{11} increases. Combined with the decrease in W_{pump} as well as the increase in $\eta_{\text{pump, is}}$, it can be seen that there will be some reduction in the exergy loss of the working fluid pump.

4.1.3. Performance of the expander

Fig. 7(a) illustrates the variation patterns of difference pressure (ΔP_{exp}) and pressure ratio (P_{rat}) of the expander with T_{11} . It can be seen that as T_{11} increases, ΔP_{exp} and P_{rat} increase monotonically in the range of 0.66–0.73 MPa and 2.68–2.86 MPa, respectively. This is because the inlet pressure of the expander increases with the increase in T_{11} . However, the pressure at the outlet of the expander increases only slightly for a constant cooling capacity. Combining Eqs. (12) and (13), it can be seen that both ΔP_{exp} and P_{rat} increase with T_{11} , but the increase is small, which is limited by the large super-heating degree. Since Gao et al. [45] found that ΔP_{exp} can be used as a predictor of power generation. Combined with the existing results, it can be seen that the power generation of the system will show a slightly increasing trend with a small increase in difference pressure.

Fig. 7(b) displays the variation patterns of the actual enthalpy

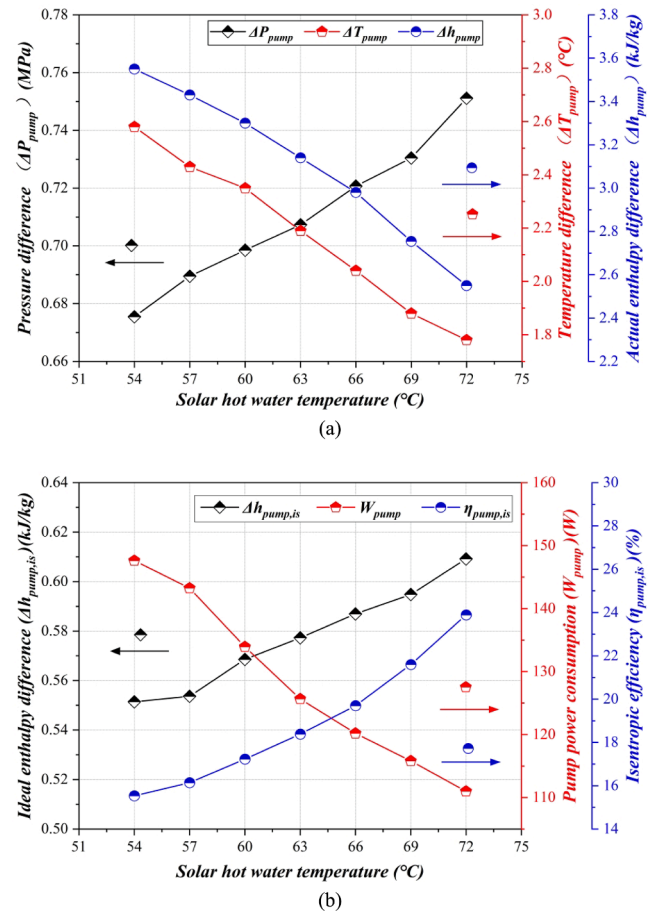


Fig. 6. Variation of working fluid pump performance with T_{11} increasing.

difference (Δh_{exp}), ideal enthalpy difference ($\Delta h_{\text{exp, is}}$) and isentropic efficiency ($\eta_{\text{exp, is}}$) of the expander with T_{11} . With the increase in T_{11} , Δh_{exp} , $\Delta h_{\text{exp, is}}$ and $\eta_{\text{exp, is}}$ monotonically increase in the ranges of 7.64–11.01 kJ/kg, 21.50–25.01 kJ/kg and 35.54–44.06 %, respectively. The reason for the increase of Δh_{exp} and $\Delta h_{\text{exp, is}}$ is that the inlet enthalpy of the expander is substantially increased with the increase of the evaporation temperature, the actual outlet enthalpy and ideal outlet enthalpy are both slightly increased. Considering that the increase in the actual outlet enthalpy is smaller than the increase in the ideal outlet enthalpy, according to Eq. (16), it can be seen that $\eta_{\text{exp, is}}$ tends to increase. The small increase in ΔP and the increase in $\eta_{\text{exp, is}}$ will lead to the reduction of the exergy loss of the expander, which will lead to the increase of the system output power and thermal efficiency.

4.1.4. Overall system performance

Fig. 8(a) illustrates the variation patterns of output power (W_{exp}), net output power (W_{net}) and power generation (W_{ele}) of the system with T_{11} . With the increase in T_{11} , W_{exp} , W_{net} and W_{ele} increase monotonically in the ranges of 346.91–475.07 W, 203.28–364.07 W and 297.87–324.09 W, respectively. According to equation (17), the change in W_{exp} depends on m_{wf} and Δh_{exp} together. The increase in Δh_{exp} is greater than the decrease in m_{wf} , so the W_{exp} continues to increase. On this basis, considering the W_{pump} varies in a small range, it can be seen that the W_{net} is constantly increasing. Additionally, the small increase in W_{ele} verifies that the pressure difference of the expander can be used as an evaluation criterion for the power generation of the system.

Fig. 8(b) presents the variation patterns of thermal efficiency (η_{th}), power generation efficiency (η_{ele}) and exergy efficiency (η_{ex}) of the system with T_{11} . As T_{11} increases, η_{th} , η_{ele} and η_{ex} increase monotonically

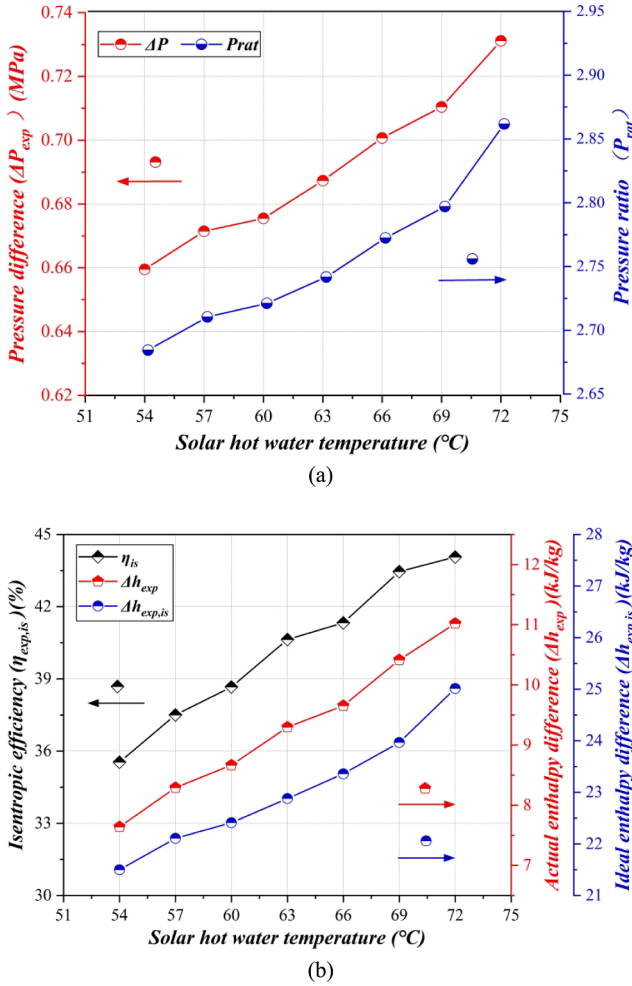


Fig. 7. Variation of expander performance with T_{11} increasing.

in the ranges of 2.02–3.26 %, 2.93–3.15 % and 37.93–41.01 %, respectively. From Eqs. (30) and (31), it can be seen that η_{th} and η_{ele} are defined in a similar way, both calculating costs in terms of the heating value released by the system, the difference being that using W_{exp} and W_{ele} respectively as benefits. Both W_{exp} and W_{ele} are greater than the increase in calorific value, so the corresponding η_{th} and η_{ele} continue to increase. For η_{ex} , its value increases with increasing T_{11} , indicating that the system is able to use the energy more efficiently and convert more heat into useful power. Considering that the system exergy efficiency is also affected by the exergy loss of each component, the amount and the percentage of exergy loss of each component are analysed below.

Fig. 9(a) displays the variation patterns of component exergy loss ($Ex_{D,i}$) and the total system exergy loss ($Ex_{D,sys}$) with T_{11} . It can be seen that $Ex_{D,sys}$ increases from 1.15 kW to 1.44 kW. The expander accounts for the largest amount of exergy loss, followed by the condenser, the superheater, the working fluid pump and the evaporator. With the increase in T_{11} , the heat transfer coefficient of the heat exchanger continues to decrease, making the heat transfer process less than optimal, resulting in increases in $Ex_{D,eva}$, $Ex_{D,sup}$ and $Ex_{D,con}$. However, the decrease in U_{eva} is small, so the change in $Ex_{D,eva}$ is not significant. In addition, the increase in the difference pressure of the expander is small, while the isentropic efficiency continues to increase, leading to an increase in the heat work capacity, and therefore $Ex_{D,exp}$ continues to decrease. The working fluid pump benefits from the increase in its isentropic efficiency and a small change in power consumption, resulting in $Ex_{D,pump}$ slight decrease.

To better understand the component exergy loss, component exergy

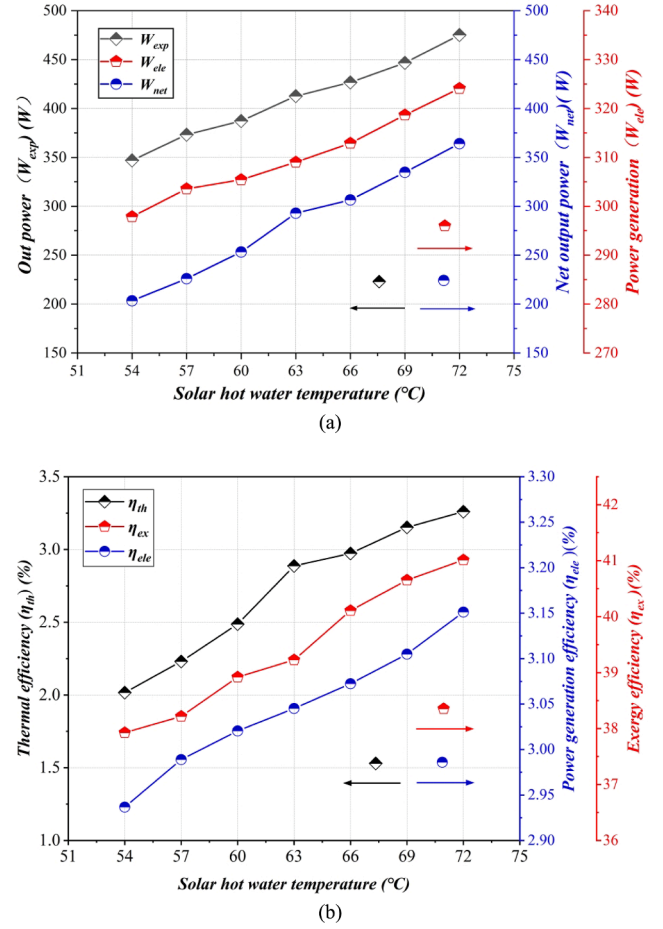


Fig. 8. Variation of the overall system performance with T_{11} increasing.

loss percentage (ω_i) is calculated and the results of the analysis are shown in Fig. 9(b). Obviously, the expander represents the largest percentage of exergy loss in the power generation system, followed by the condenser and the superheater. With the increase in T_{11} , ω_{exp} decreases from 56 % to 39 %, ω_{con} and ω_{sup} increase in the range of 18–28 % and 10–23 %, respectively. The reason why the expander accounts for the most exergy loss is that it corresponds to a low isentropic efficiency, resulting in a much lower work capacity than desirable. Additionally, ω_{con} is larger than ω_{sup} under the same conditions because the experiments were carried out during the summer when the ambient temperature was as high as 35 $^{\circ}C$, which increased the exergy loss of condenser to some extent.

From the above, it can be seen that as T_{11} increases, the superheating degree of the system increases, causing the heat transfer process in the heat exchanger to be less than ideal and causing more exergy losses. Therefore, in the following section, the super-heating degree will be reduced by changing the working fluid flow rate to observe the working effect of the components and the performance of the system.

4.2. Effect of working fluid flow rate on system performance

During the experiments, only the working fluid pump frequency was varied to increase the working fluid flow rate (m_{wf}) in the range of 124.33–216.73 kg/h. Potential optimisation directions are further analysed by investigating the effect of m_{wf} on the components as well as the system performance. The main experimental operating parameters are presented in Table 10.

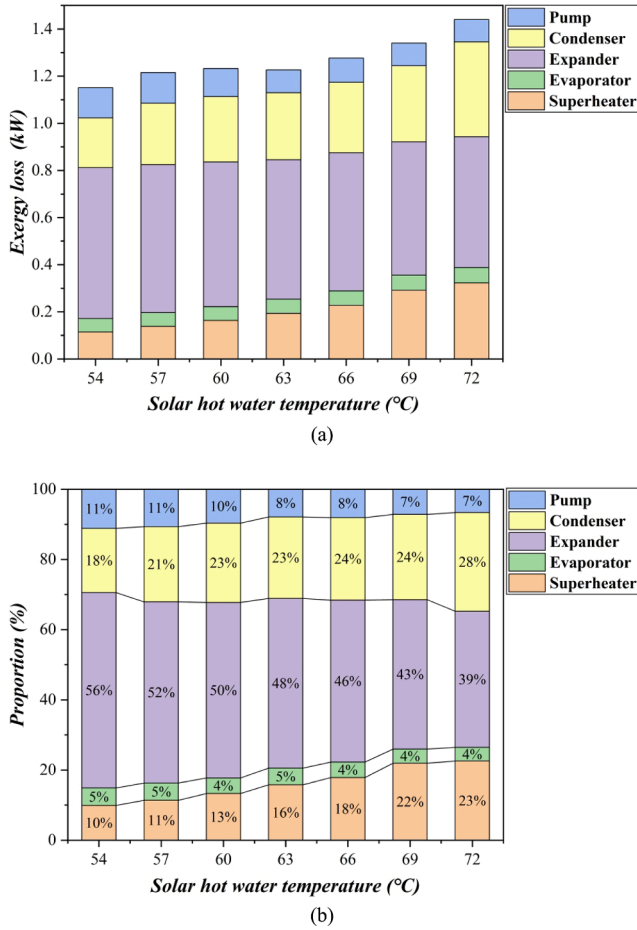


Fig. 9. The exergy loss and exergy loss proportion of the components with T_{11} increasing.

Table 10

The main operating parameters used in the experiment.

Parameters	Value	Unit
Environmental temperature	35	°C
Temperature of warm seawater at evaporator inlet	31	°C
Temperature of cold seawater at condenser inlet	5	°C
Temperature of solar hot water at collector outlet	64	°C
The Surface warm seawater flow rate	1.5	m ³ /h
The deep cold seawater flow rate	1.5	m ³ /h
The solar hot water flow rate	1.5	m ³ /h
The working fluid flow rate	124.33–216.73	kg/h

4.2.1. Performance of the heat exchanger

Fig. 10 presents the variation patterns of temperature difference (ΔT) and energy released (Q) with m_{wf} on the water side in the heat exchanger, respectively. With the increase in m_{wf} , ΔT_{eva} and ΔT_{sup} monotonically increase in the ranges of 0.75–0.83 °C, 4.80–7.95 °C, $Q_{eva,water}$ and $Q_{sup,water}$ monotonically increase in the ranges of 1.21–1.46 kW and 6.01–9.41 kW, respectively. This is due to the fact that the inlet temperature of the heat source in the evaporator and superheater remains constant and the increase in m_{wf} removes more heat. It causes the outlet temperature of the heat exchanger to decrease, resulting in an increase in the temperature difference of the heat source and an increase in the calorific value released. However, ΔT_{con} and $Q_{con,water}$ have a higher increase. This is due to the fact that the larger m_{wf} leads the working fluid to only partially phase change under the heating effect of 31 °C in the evaporator and then enters the superheater to exchange heat with a higher temperature heat source to complete the full phase

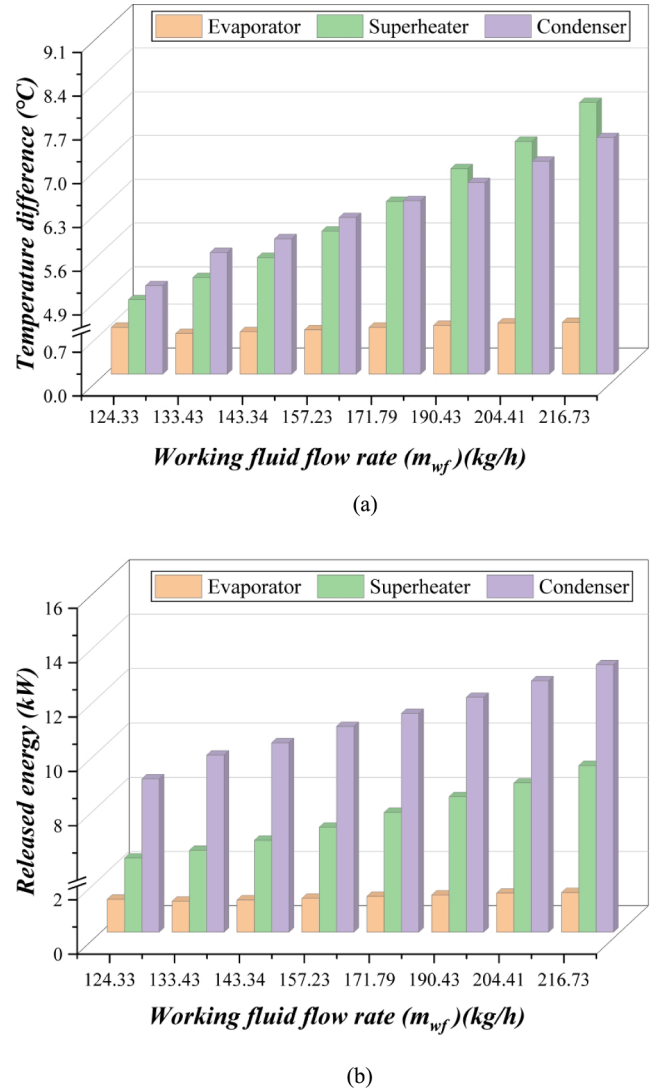


Fig. 10. Variation of temperature difference and released energy of the water in the heat exchanger with m_{wf} increasing.

change. In addition, ΔT_{con} and $Q_{con,water}$ increase monotonically in the range of 5.02–7.39 °C and 8.92–13.12 kW, respectively. This is due to the fact that the cold water inlet temperature is maintained at 4 °C and the increase in m_{wf} takes more cold away, causing the cold water outlet temperature to rise, resulting in an increase in the temperature difference and an increase in the amount of cold released.

Fig. 11 illustrates the variation patterns of $LMTD$ and U with m_{wf} . With the increase of m_{wf} , $LMTD_{eva}$, $LMTD_{sup}$ and $LMTD_{con}$ decrease monotonically in the ranges of 7.21–6.66 °C, 10.71–9.71 °C and 8.86–6.85 °C, U_{eva} , U_{sup} and U_{con} monotonically decrease in the ranges 56.05–71.86 W/(m²·K), 186.93–323.33 W/(m²·K) and 154.78–294.77 W/(m²·K), respectively. The reason for the constant increase in U is that the increase in m_{wf} increases the heat value released from the heat source and the amount of cold released from the cold source, which causes the evaporation rate and condensation rate of the working fluid to accelerate. On the other hand, it is benefiting from the declining $LMTD$. This phenomenon confirms that the variation of the heat transfer coefficient is a combined effect of the heat transfer rate and the $LMTD$. This means that the improvement of the heat transfer effect in the heat exchanger will reduce the exergy losses to some extent. In addition, the increase of U_{eva} is small compared to U_{sup} and U_{con} , only 15.81 W/(m²·K), which shows that the corresponding heat transfer enhancement

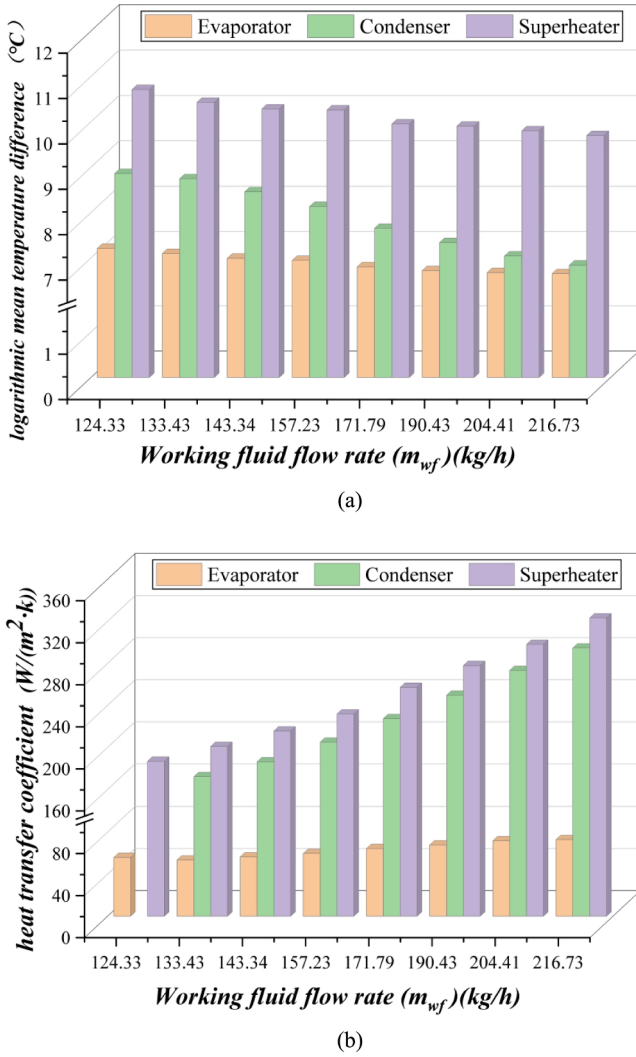


Fig. 11. Variation of the heat transfer effect of heat exchangers with m_{wf} increasing.

effect in the evaporator is relatively weak.

Fig. 12 displays the variation pattern of P_{10} , P_4 and ΔT_{Super} with m_{wf} . With the increase in m_{wf} , P_{10} and P_4 increase monotonically in the ranges of 0.91–1.27 MPa and 0.38–0.44 MPa, respectively. The reason for the increase in P_{10} is that the increase in m_{wf} causes the working fluid

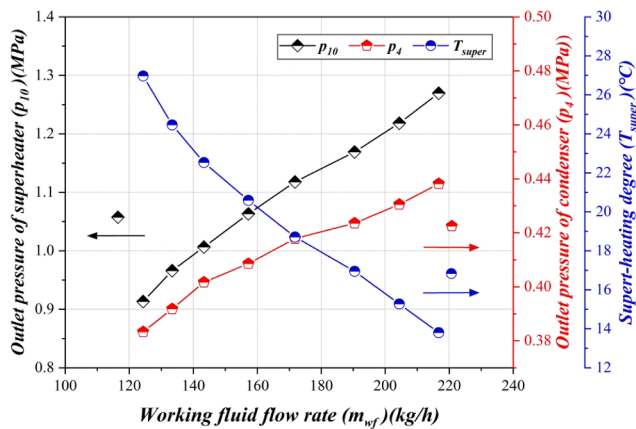


Fig. 12. Variation of outlet pressure and superheat of heat exchanger with m_{wf} increasing.

in the superheater to undergo sufficient phase change, resulting in the gradual decrease of the liquid phase zone in the superheater and the increase of the gas phase zone. The increase in P_{10} causes the outlet pressure of the expander to increase slightly. Considering that the trend of P_4 and the outlet pressure of the expander are very similar, the trend of P_4 is also increasing. In addition, since the temperature at the outlet of the working fluid in the superheater is basically unchanged, and the increase in P_{10} causes the saturation temperature of the working fluid to increase continuously, combined with Eq. (9), it can be seen that ΔT_{Super} decreases continuously. When m_{wf} is 124.33 kg/h and 216.73 kg/h, the maximum and minimum values of ΔT_{Super} are 26.97 $^{\circ}\text{C}$ and 13.80 $^{\circ}\text{C}$, respectively.

4.2.2. Performance of the working fluid pump

Fig. 13 illustrates the effect of m_{wf} on the performance of the working fluid pump. In Fig. 12(a), it can be seen that as m_{wf} increases, ΔT_{pump} decreases from 3.39 $^{\circ}\text{C}$ to 1.89 $^{\circ}\text{C}$ and ΔP_{pump} increases from 0.55 MPa to 0.91 MPa. It is the decrease in ΔT_{pump} and the increase in ΔP_{pump} that causes Δh_{pump} to decrease. As shown in the Fig. 13(a), Δh_{pump} decreases from 4.70 kJ/kg to 2.74 kJ/kg.

In addition, Δh_{pump} increases in the small range of 0.39–0.66 kJ/kg, which combined with the decrease in Δh_{pump} results in an increasing $\eta_{pump, is}$ from Eq. (28). As shown in Fig. 13(b), $\eta_{pump, is}$ increases from 8.29 % to 24.09 %. In addition, W_{pump} fluctuates in the range of 161.55–170.07 W with less variation. This is because the variation patterns of W_{pump} is the combined effect of m_{wf} and Δh_{pump} . The small fluctuation of W_{pump} and the increase of $\eta_{pump, is}$ will result in the

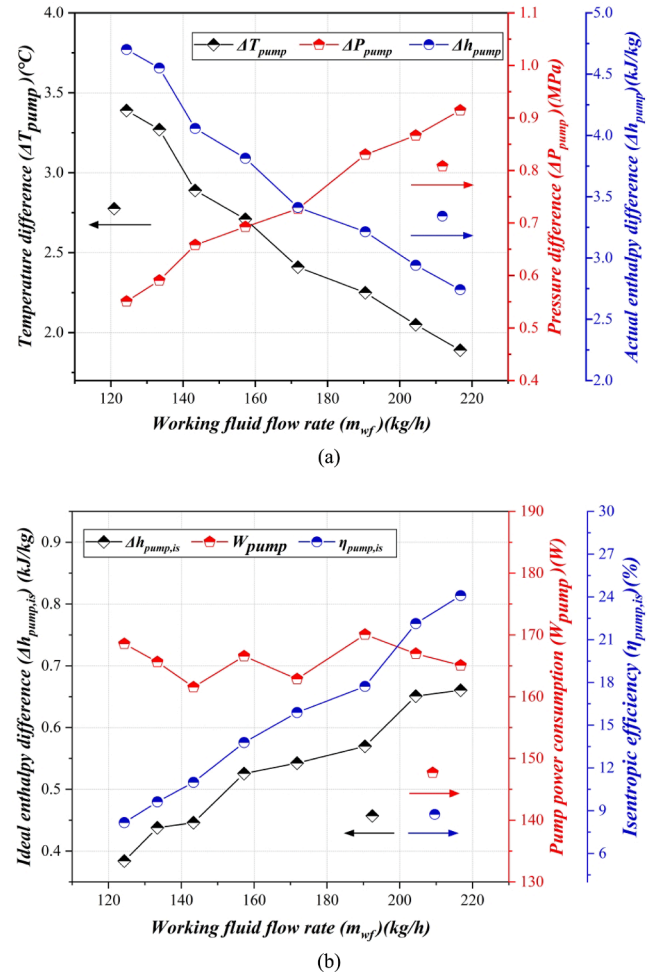


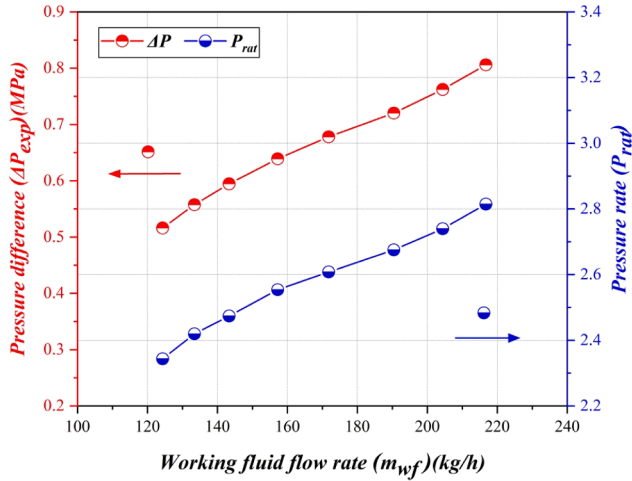
Fig. 13. Variation of working fluid pump performance with m_{wf} increasing.

reduction of the exergy loss of the working fluid pump.

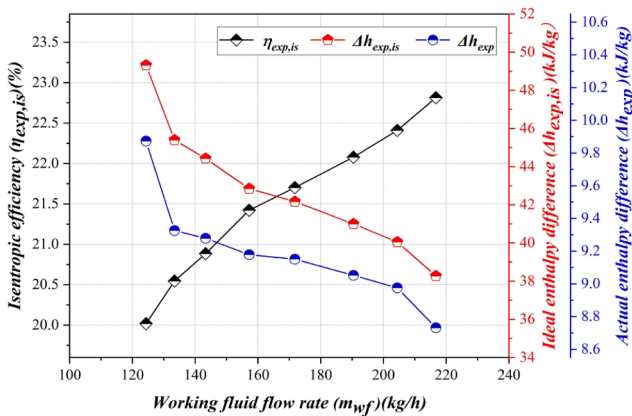
4.2.3. Performance of the expander

Fig. 14(a) illustrates the variation patterns of ΔP_{exp} and P_{rat} with m_{wf} . With the increase in m_{wf} , the ΔP_{exp} and P_{rat} increase monotonically in the ranges of 0.52–0.81 MPa and 2.34–2.74 MPa, respectively. This is because as m_{wf} increases, the inlet pressure of the expander and the outlet pressure of the superheater have a similar increasing trend. In addition, the pressure at the outlet of the expander increases only slightly for a constant cooling capacity. Combining Eqs. (13) and (14) gives an increase in ΔP_{exp} and P_{rat} .

Fig. 14(b) presents the variation patterns of Δh_{exp} , $\Delta h_{exp,is}$ and $\eta_{exp,is}$ with m_{wf} . As m_{wf} increases, the Δh_{exp} and $\Delta h_{exp,is}$ decrease monotonically in the ranges 9.87–8.73 kJ/kg and 49.32–38.27 kJ/kg, respectively. This is due to the fact that the inlet temperature of the expander increases less while the inlet pressure increases significantly, causing both the inlet enthalpy and the ideal outlet enthalpy to decrease continuously. However, the ideal outlet enthalpy decreases more resulting in a continuous increase in $\Delta h_{exp,is}$. In addition, the outlet temperature of the expander decreases by a larger amount while the increase in outlet pressure is smaller, resulting in a decrease in outlet enthalpy. Nevertheless, the decrease in the outlet enthalpy is larger than that of the inlet enthalpy, resulting in a constant decrease in Δh_{exp} . Considering that the decrease in Δh_{exp} is larger than $\Delta h_{exp,is}$, combined with Eq. (16), it can be seen that $\eta_{exp,is}$ increases. As shown in the Fig. 13(b), when m_{wf} is 124.33 kg/h and 216.73 kg/h, the minimum and maximum values of $\eta_{exp,is}$ are 20.02 % and 22.81 %, respectively.



(a)



(b)

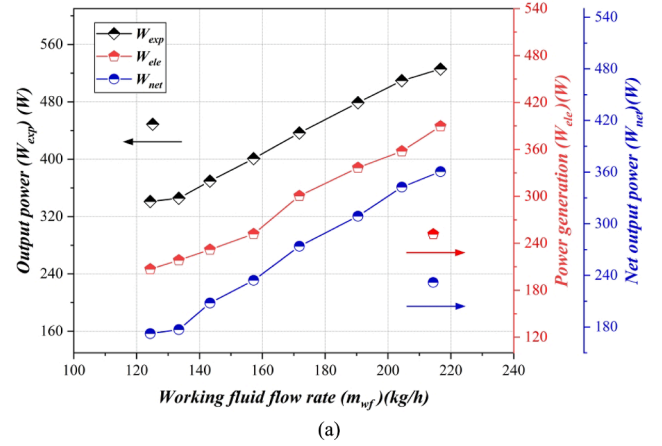
Fig. 14. Variation of expander performance with increasing.

4.2.4. Overall performance of the system

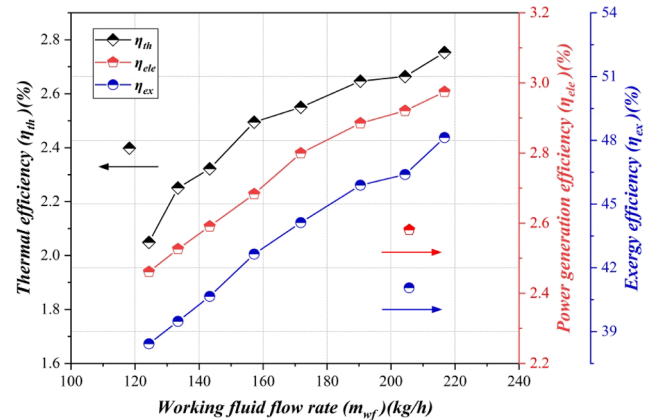
Fig. 15(a) presents the variation patterns of W_{exp} , W_{net} and W_{ele} with m_{wf} . As m_{wf} increases, the W_{exp} , W_{net} and W_{ele} increase monotonically in the ranges of 340.99–525.67 W, 172.32–360.61 W and 207.01–389.65 W, respectively. The increase in m_{wf} causes the hot water to release its calorific value faster, resulting in faster evaporation of the working fluid, which increases the flow rate and flow of vapor, which in turn increases the speed of the turbine, so that W_{exp} increases continuously. On this basis, since the fluctuation range of W_{pump} is small, combined with Eq. 29, it can be seen that W_{net} is constantly increasing. In addition, there is a large pressure difference at the inlet and outlet of the expander, which leads to a constant increase in W_{ele} , where the maximum power generation reaches 389.65 W.

Fig. 15(b) illustrates the variation patterns of η_{th} , η_{ele} and η_{ex} with m_{wf} . With the increase of m_{wf} , the η_{th} , η_{ele} and η_{ex} increase monotonically in the range of 2.05–2.75 %, 2.46–2.97 % and 38.01–48.13 %, respectively. Due to the growth of W_{net} and W_{ele} is higher than the growth of heat input, and therefore η_{th} and η_{ele} show a similar increasing trend. In addition, the increase in η_{ex} indicates that the growth rate of net power output is higher than the rate of heat input loss, resulting in higher exergy utilisation.

Fig. 16(a) demonstrates the variation patterns of the component exergy loss ($Ex_{D,i}$) and the total system exergy loss ($Ex_{D,sys}$) with m_{wf} . It can be seen that $Ex_{D,sys}$ increases from 1.03 kW to 1.36 kW. With the increase of m_{wf} , the heat transfer temperature difference in the condenser and the superheater continues to increase, as does the degree of turbulence in the circulating water. This leads to an increase in the heat transfer coefficient, which increases the heat transfer effect, so



(a)



(b)

Fig. 15. Variation of overall system performance with m_{wf} increasing. (a) Power, (b) Efficiency.

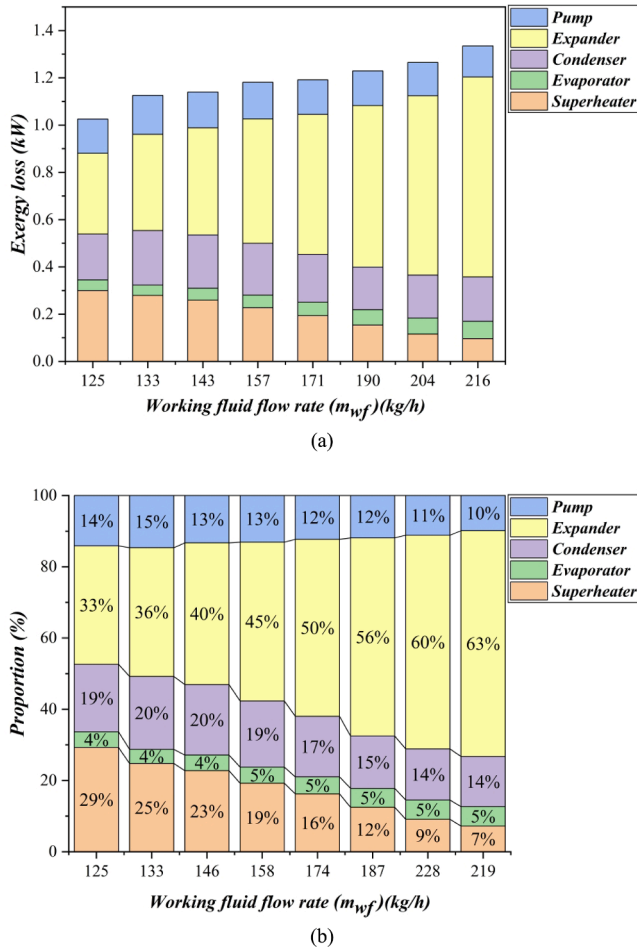


Fig. 16. The exergy loss and exergy loss proportion of the components with m_{wf} increasing.

$Ex_{D,con}$ and $Ex_{D,sup}$ gradually decrease. In contrast, $Ex_{D,eva}$ increases slightly due to the larger m_{wf} and the fact that the temperature of the hot water in the evaporator is only 31 °C, which causes the working fluid to leave the evaporator almost before the phase change occurs, resulting in a smaller increase in U_{eva} , leading to an insignificant improvement in the heat transfer effect. Additionally, the gas flow rate in the expander increases, which intensifies the exergy loss and causes $\eta_{exp,is}$ to decrease, leading to a significant increase in $Ex_{D,sup}$.

To better understand the component exergy loss, component exergy loss percentage (ω_i) is calculated and the results of the analysis are shown in Fig. 16(b). As m_{wf} increases, the expander accounts for the largest amount of exergy loss in the power generation system, and ω_{exp} increases from 33 % to 63 %. This is due to the fact that the expander not only has a low isentropic efficiency, but also its maximum inlet pressure is only 1.38 MPa, which imposes some limitations on the increase in m_{wf} , resulting in its working capacity cannot be realised to a large extent. In addition, increasing m_{wf} reduces the super-heating degree of the system, improves the utilisation of heat and cold sources, and reduces the irreversible loss in the process of evaporation and condensation. Therefore, ω_{sup} and ω_{con} constantly decrease, with corresponding variations in the range of 29–7 % and 19–14 %, respectively. Additionally, The changes in ω_{eva} and ω_{pump} are smaller, corresponding to a range of 4–5 % and 14–10 %.

4.3. Comparison between S-OTEC and OTEC

In the experimental process, the OTEC system was first controlled to

reach the operating limit by combining the magnitude of super-heating degree and the pressure capacity of the expander. Then the solar heating cycle was introduced to obtain the S-OTEC system, and finally the collector outlet temperature was stabilised at 64 °C, maintaining the same experimental parameters (seawater temperature and flow, working fluid flow rate and ambient temperature) as those of the OTEC system. Under the above conditions, the performance differences between the OTEC and S-OTEC systems are compared. The main experimental operating parameters are given in Table 11.

4.3.1. Performance of the expander

Fig. 17(a) illustrates the difference of P_1 , P_2 , ΔP_{exp} and P_{rat} between S-OTEC and OTEC. It can be seen that the S-OTEC system has higher P_1 (0.85 MPa vs. 0.71 MPa), slightly higher P_2 (0.39 MPa vs. 0.37 MPa), higher ΔP_{exp} (0.46 MPa vs. 0.34 MPa) and P_{rat} (2.17 vs. 1.92). The P_1 , P_2 , ΔP_{exp} and P_{rat} are approximately improved by 18.06 %, 5.41 %, 35.29 % and 13.02 %, respectively. This is due to the fact that the S-OTEC system benefits from solar energy, which allows the circulating working fluid to reach higher temperatures and pressures at the inlet of the expander. In addition, under constant cold end conditions, the outlet pressure increases only slightly as the inlet pressure increases. Therefore, the S-OTEC system achieves higher ΔP_{exp} and P_{rat} .

Fig. 17(b) displays the difference of h_1 , h_2 , Δh_{exp} and $\eta_{exp,is}$ between S-OTEC and OTEC. It can be seen that the S-OTEC system has higher h_1 (446.41 kJ/kg vs. 416.66 kJ/kg), higher h_2 (436.75 kJ/kg vs. 412.61 kJ/kg), higher Δh_{exp} (9.66 kJ/kg vs. 4.05 kJ/kg), higher $\eta_{exp,is}$ (52.67 % vs. 29.93 %). The Δh_{exp} and $\eta_{exp,is}$ increase by approximately 137.68 % and 75.98 %, respectively. This is due to the fact that the S-OTEC system benefited from the support of solar energy, which resulted in a significant increase in the inlet temperature of the expander, leading to an increase in h_1 and h_2 , but a higher increase in h_1 resulted in a higher Δh_{exp} and $\eta_{exp,is}$.

4.3.2. Overall performance of the system

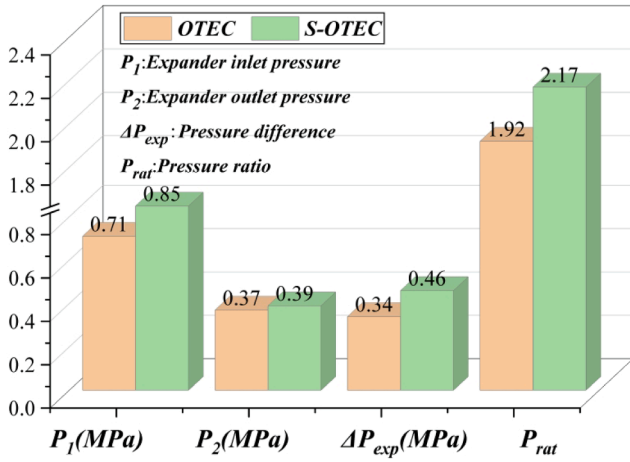
Fig. 18(a) shows the difference in power performances (W_{exp} , W_{pump} , W_{net} , W_{ele}) between S-OTEC and OTEC. It can be seen that the S-OTEC system has higher W_{exp} (127.14 W vs. 306.89 W), higher W_{pump} (75.28 W vs. 156.88 W), higher W_{net} (51.86 W vs. 150.01 W) and higher W_{ele} (96.85 W vs. 156.31 W). The W_{exp} , W_{net} and W_{ele} increase by approximately 141.38 %, 189.26 % and 61.39 %, respectively. The higher enthalpy difference achieved by the expander in the S-OTEC system results in a higher W_{exp} for the same working fluid flow rate. In addition, the increase in power output of the S-OTEC system compared to the OTEC system is significantly higher than the increase in pump power consumption, giving the S-OTEC system a higher W_{net} . The higher W_{ele} in the S-OTEC system benefits from its higher difference pressure.

Fig. 18(b) presents the difference in system efficiency (η_{th} , η_{ele} , η_{ex}) between S-OTEC and OTEC. It can be seen that the S-OTEC system has a higher η_{th} (2.06 % vs. 0.805 %), higher η_{ele} (2.09 % vs. 1.50 %), higher η_{ex} (41.66 % vs. 32.58 %). The η_{th} , η_{ele} and η_{ex} increase by approximately 155.90 %, 39.33 % and 27.87 %, respectively. Although the S-OTEC system has a higher heat input compared to the OTEC system, the S-

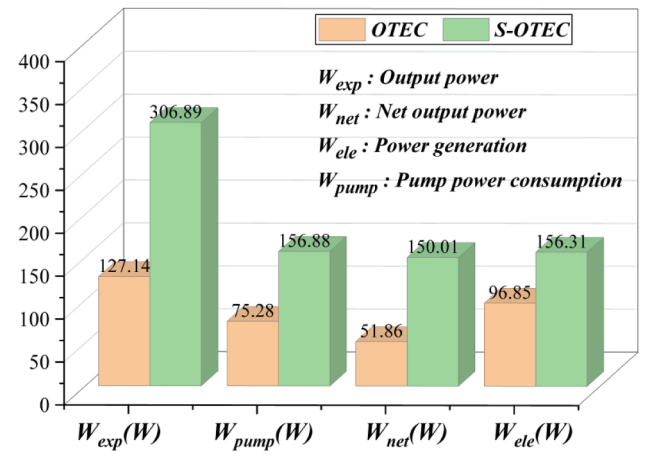
Table 11

The main operating parameters used in the experiment.

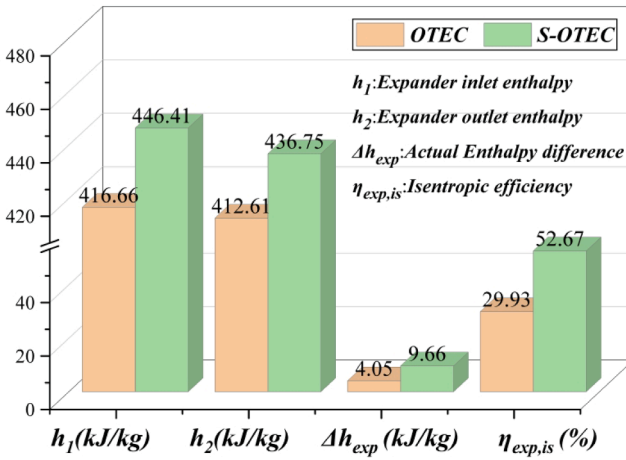
Parameters	Unit	Value	
		OTEC	S-OTEC
Environmental temperature	°C	35	35
Temperature of warm seawater at evaporator inlet	°C	31	31
Temperature of cold seawater at condenser inlet	°C	5	5
Temperature of solar hot water at collector outlet	°C	/	64
The warm seawater flow rate	m ³ /h	2.5	2.5
The cold seawater flow rate	m ³ /h	2.5	2.5
The solar hot water flow rate	m ³ /h	/	2.5
The working fluid flow rate	m ³ /h	112.77	112.77



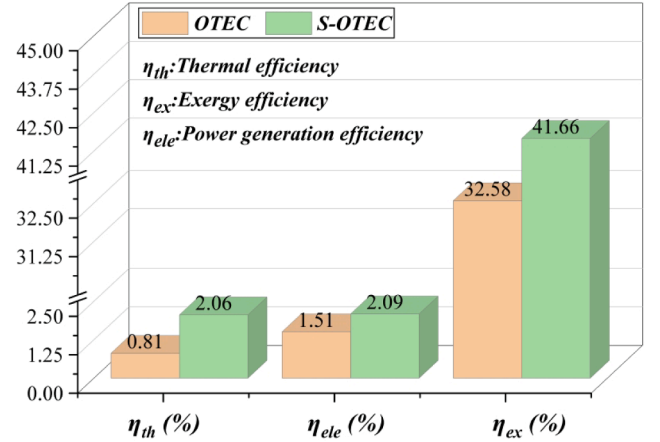
(a)



(a)



(b)



(b)

Fig. 18. Comparison of system performance in S-OTEC and OTEC.

Fig. 17. Comparison of expander performance in S-OTEC and OTEC systems.

OTEC system has the higher W_{net} and W_{ele} , and the increase in both W_{net} and W_{ele} is greater than the increase in heat input, so the S-OTEC system has a higher η_{th} and η_{ele} . In addition, the higher η_{ex} indicates that the S-OTEC system has a higher exergy conversion rate.

5. Conclusion

In this study, a 1 kW Solar-Ocean Thermal Energy Conversion system with R134a as the working medium was established and experimental setup was carried out. The effects of solar hot water temperature and working fluid flow rate on the performance of the main working components and system were investigated using energy analysis and exergy analysis. Moreover, differences in system performance between the S-OTEC and OTEC were compared. The main conclusions are as follows:

1) In the S-OTEC system, the heat transfer coefficient of the heat exchanger decreases and the isentropic efficiency of the working fluid pump and expander increase as the solar hot water temperature increases; the heat transfer coefficient of the heat exchanger and the isentropic efficiencies of the working fluid pump and expander increase as the working fluid flow rate increases.

2) Increasing the solar hot water temperature or working fluid flow rate will improve the performance of the S-OTEC system. When the solar hot water temperature increases from 54 °C to 72 °C, the thermal efficiency and exergy efficiency of the system increase in the ranges of

2.02–3.26 % and 37.93–41.01 %, respectively; when the working fluid flow rate increases from 124.33 kg/h to 216.73 kg/h, the thermal efficiency and exergy efficiency of the system increase in the ranges of 2.05–2.75 % and 38.01–48.13 %, respectively.

3) In the S-OTEC system, as the solar hot water temperature increases, the total exergy losses of the system increase in the range of 1.15–1.44 kW; as the working fluid flow rate increases, the total exergy losses of the system increase in the range of 1.03–1.36 kW. Additional, the percentage of exergy loss of the expander is consistently the highest of all components. Therefore, optimizing the expander and its operating parameters is a potential way to further improve the system performance.

4) For the same working fluid flow rate and seawater temperature difference, the thermal efficiency, power generation efficiency and energy efficiency of the S-OTEC system were approximately 154.32 %, 39.33 % and 27.87 % higher than those of the OTEC system, respectively.

Considering that solar energy has the property of fluctuating energy, it is expected that future research will be carried out on the dynamic analysis of solar energy for the S-OTEC system. The performance of the experimental system will be studied in terms of energy, exergy and economic cost, and it will be analysed whether the incorporation of solar energy can always be beneficial to improve the performance of the OTEC system.

CRediT authorship contribution statement

Wenzhong Gao: Writing – original draft, Project administration, Funding acquisition, Conceptualization. **Fei Wang:** Writing – review & editing, Writing – original draft, Visualization, Data curation, Conceptualization. **Zhen Tian:** Writing – review & editing, Writing – original draft, Methodology. **Yuan Zhang:** Writing – review & editing, Writing – original draft, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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